

Project Initialization and Planning Phase

Date	04 July 2026
Team ID	
Project Title	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to develop an AI-powered voice-based concept understanding analyser that evaluates a user's explanation of a concept through speech. The system uses speech recognition, natural language processing, and audio analysis techniques to measure conceptual accuracy, speaking quality, and provide detailed feedback reports.

The solution helps learners improve communication skills, identify knowledge gaps, and enhance their understanding through automated AI-based evaluation.

Project Overview	
Objective	To create an AI-based system that analyzes spoken explanations, evaluates concept understanding, and provides instant feedback with performance reports.
Scope	The project includes voice input processing, speech-to-text conversion, semantic similarity analysis, audio quality evaluation, score generation, and report visualization using a web application.
Problem Statement	
Description	Students often lack immediate feedback on their conceptual understanding and explanation skills. Manual evaluation takes time and may vary between evaluators.
Impact	The solution improves self-learning by providing quick, consistent, and AI-generated feedback, helping users enhance knowledge clarity and communication skills.
Proposed Solution	
Approach	Use AI and NLP techniques to convert speech into text, compare explanations with reference concepts, analyze audio features, and generate performance scores.
Key Features	- Voice recording and audio upload support. - Speech-to-text conversion using AI models.

	- Semantic similarity-based concept evaluation. - Audio quality and fluency analysis. - Interactive dashboard with feedback reports. - PDF report generation.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel i5 CPU / Cloud CPU Environment
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	Minimum 10 GB Storage
Software		
Frameworks	Python frameworks	Streamlit Framework
Libraries	Additional libraries	Whisper, Sentence Transformers, Librosa, Pandas, NumPy, Matplotlib, ReportLab
Development Environment	IDE	Visual Studio Code
Data		
Data	Source, size, format	User audio input files (.wav/.mp3), Reference concept text data, Generated reports (PDF)